

Function Call Stack & Recursion

Stack

Stack is a data structure that stores items in an Last-In/First-Out manner.



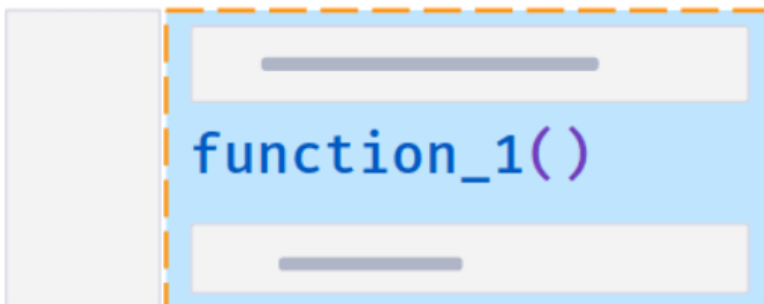
Calling a Function

Calling `function_1()` inside `function_2()`

```
def function_1():
```



```
def function_2():
```



Code

```
2     len_list = len(list_x)
3     for i in range(len_list):
4         x = list_x[i]
5         list_x[i] = x * x
6     largest = max(list_x)
7     return largest
8
9 list_a = [1,-3,2]
10 result = get_largest_sqr(list_a)
11 print(result)
```

PYTHON

Expand ▼

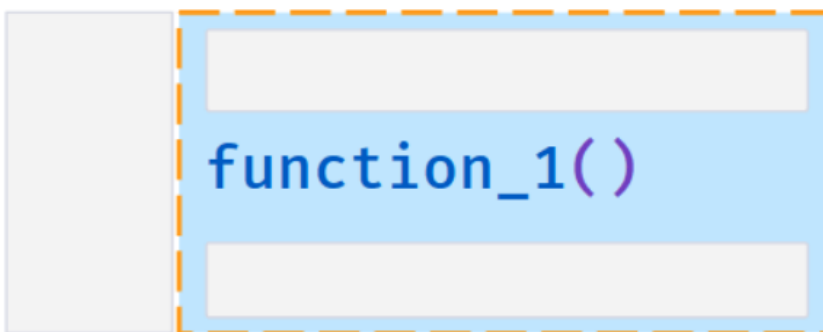
Output

9

Recursion

A function calling itself is called a **Recursion**

```
def function_1():
```



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Let's understand recursion with a simple example of multiplying N numbers

Multiply N Numbers

PYTHON

```
1 def factorial(n): # Recursive Function
2     if n == 1: # Base Case
3         return 1
4     return n * factorial(n - 1) # Recursion
5 num = int(input())
6 result = factorial(num)
7 print(result)
```

Base Case

A recursive function terminates when base condition is met

Input

3

Output

6

Without Base case

Code

PYTHON

```
1 def factorial(n):
2     return n * factorial(n - 1)
3 num = int(input())
4 result = factorial(num)
5 print(result)
```

Input

3

Output

```
RecursionError: maximum recursion depth exceeded
```

Identify the Mistake - Sum of First N Natural Numb



● Solved

Helpful

Write a program to recursively compute the sum of the numbers from 1 to N.

Input

The input will be a single line containing a positive integer (N).

Output

The output should be a single line containing the sum of the first N natural numbers.

Explanation

For example, if the given N is 4, your code should print the sum of numbers from 1 to 4. So the output should be 10.

Sample Input 1

4

Sample Output 1

10

Sample Input 2

5

Sample Output 2

15

Sum of Numbers



Easy

● Solved

Helpful

A function is given in the prefilled code that takes a list `L` as an argument.

Write a program that returns the sum of the numbers in the list `L` using Recursion.

Input

The input will be a single line containing space-separated integers.

Output

The output should be a single line containing an integer that is the sum of the numbers in the list `L` that is passed as an argument.

Explanation

For example, if the given space-separated numbers are `2 1 5 6`,

The output should be `14`.

Sample Input 1

```
2 1 5 6
```

Sample Output 1

```
14
```

Sample Input 2

```
-10 12 1 3 4 30 46 97 78
```

Sample Output 2

```
261
```

Factorial - 2

● Solved

Helpful



Write a program to print the factorial of N . Factorial is the product of all positive integers less than or equal to N .

Note

The factorial of 0 is 1.

Input

The input will be a single line containing a positive integer representing N .

Output

The output should be a single line containing the factorial of the given number N .

Explanation

For example, if the given number is $N = 4$,

- The factorial of 4 is 24. ($4 \times 3 \times 2 \times 1 = 24$)

So, the output should be 24.

Sample Input 1

4

Sample Output 1

24

Sample Input 2

0

Sample Output 2

1

Sum of Squares of Numbers



Easy

● Solved

Helpful

A function is given in the prefilled code that takes a list `L` as an argument.

Write a program that returns the sum of the squares of the numbers in the list `L` using Recursion.

Input

The input will be a single line containing space-separated integers.

Output

The output should be a single line containing an integer that is the sum of the squares of the numbers in the list `L` that is passed as an argument.

For example, if the given space-separated numbers are `5 2 4` ,

- The squares of the given numbers are `25, 4, 16`.
- The sum of the squares of the given numbers is `45` (`25 + 4 + 16 = 45`).

The output should be `45`.

Sample Input 1

```
5 2 4
```

Sample Output 1

```
45
```

Reverse the String - 2



Easy

● Solved

Helpful

A function is given in the prefilled code that takes a string `S` as an argument.

Write a program that returns the reverse of the given string `S` using Recursion.

Input

The input will be a single line containing a string representing `S`.

Output

The output should be a single line containing the reverse of the string `S` that is passed as an argument to the function.

Explanation

For example, if the given string is `S = "react"`,

- The characters in `react` are `r`, `e`, `a`, `c`, and `t`.
- The reverse of the string `react` is `tcaer`.

The output should be `tcaer`.

Sample Input 1

```
react
```

Sample Output 1

```
tcaer
```

Sample Input 2

```
BootStrap
```

Sample Output 2

```
partStooB
```